PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: Etch: Dicing with Laser-cutter

Requestor: **Payton Broaddus** **Date: 14/2/2019**

Non-SNF staff, please complete the following:

Badger ID: **Broaddus**

Company name/PI: Olav Solgaard

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups and the Stanford Chemical Safety Data Base sections on this website to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

N/A

2. **Vendor/manufacturer info:** address and phone number, website URL.

N/A

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

I want to use the ExFab Laser Cutter to laser back scribe a wafer, dice it, and take the pieces back to processing in a "Clean" tool without any further chemical clean. This comes from wanting to do a parameter search of etch recipes and tools to reduce LER of the photoresist (PR) used as the mask for a Si3N4 on Tox waveguide, and I don't want to waste a wafer per etch data point. Previous PROM request have dealt with dicing wafers and cleaning them to allow them to be processed in clean tools. Wafersawing a wafer puts the dies into the contaminated group, but you can do chemical cleaning steps to move it back into clean group. The mask on my wafer will be PR, so I can't do the normal coat-in-PR-and-wafersaw approach since the cleaning steps to remove the top layer of PR will also remove the underlying PR waveguide mask. Furthermore, subsequent chemical cleaning steps will surely remove any PR on the wafer. Without cleaning, I can't move the wafer dies I have created back into the clean group. The only other options available to me are using mechanical backscribes, such as a diamond

tip or the FlipScribe in the SNSF. However, tip scribing by hand tends to be non-exact in terms of dimensions, and the flipscribe requires the wafer to lie flat on its face, something that would potentially damage the

Also, I want to create an alternative to dicing on the Wafersaw which is less mechanically stressful on MEMS while still having semi-consistent dimensions. This is related to this PROM request insofar as it describes the steps to simply laser-dice a wafer.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that f the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

0) Produce Laser Scribe Wafer Holder

a) Buy 3 Acrylic Plates from Product Realization Lab Room 36 (PRL Room 36)

b) Cover in tape to reduce particle re-attachment during cutting (applied at PRL Room 36)

c) Cut in 2 of three in laser cutter: First layer no cut, Second Layer 96mm circle (~1.5mm lip wafer will sit on), Third Layer 100mm Wafer shape

   i) CO2 Laser, 10% Speed, 100% Frequency, 100% Power

d) Remove tape from plates

e) While wearing gloves Stack and attach with acrylic cement: small amount applied to where plates meet pulls the cement in between the plates via capillary action (applied at PRL Room 36)

f) rinse in DI water for 4 mins @ wbflexcore

g) Blow dry with N2 Gun

1) wbclean standard RCA clean

2) coat with 3 um of oxide in Themco1 (<u>save some as carrier wafers</u>)

3) coat with 400nm of Si3N4 in ThermcoNitride

4) YES oven

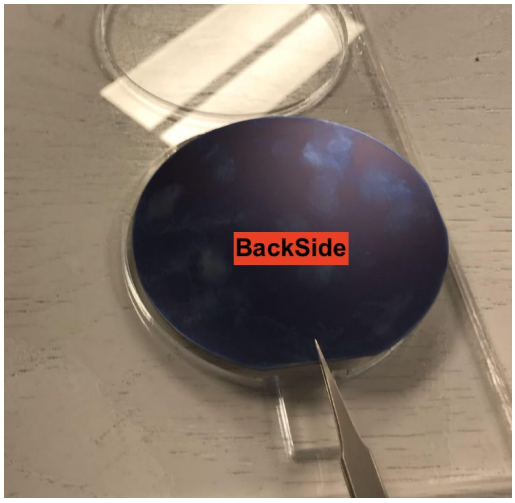
3) Coated 1um of 3615 resist. 5mm EBR. No VP (since YES oven)

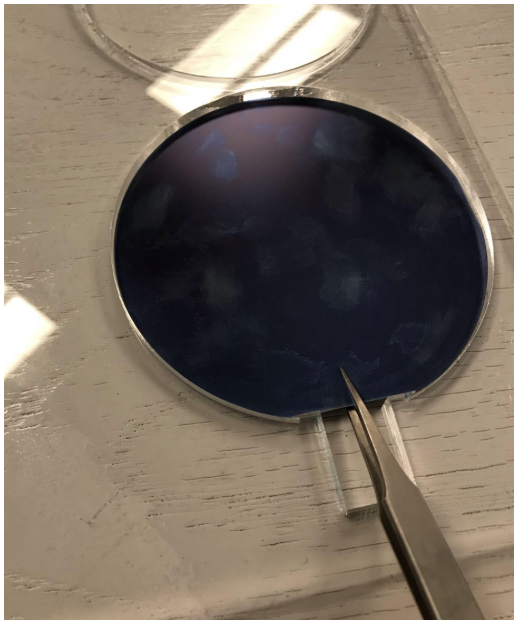
4) ASML pattern or Heidelberg pattern

5) svgdev2

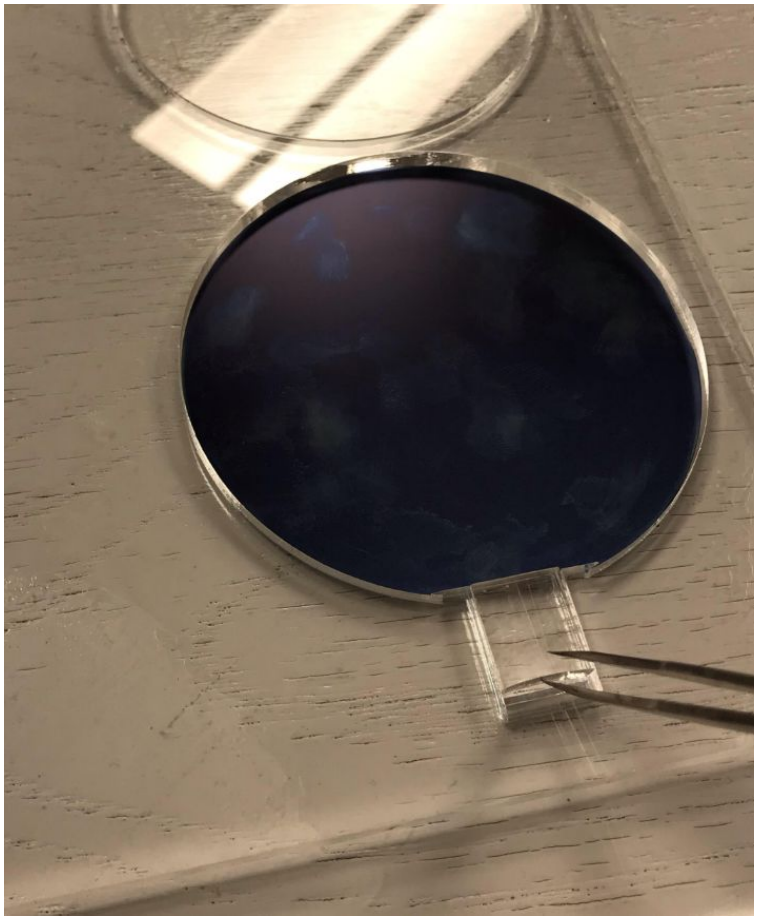
6) Transport to ExFab room: Venice in clean container

7) Place wafer with wafer face facing down in Laser Scribe Wafer Holder (see below)



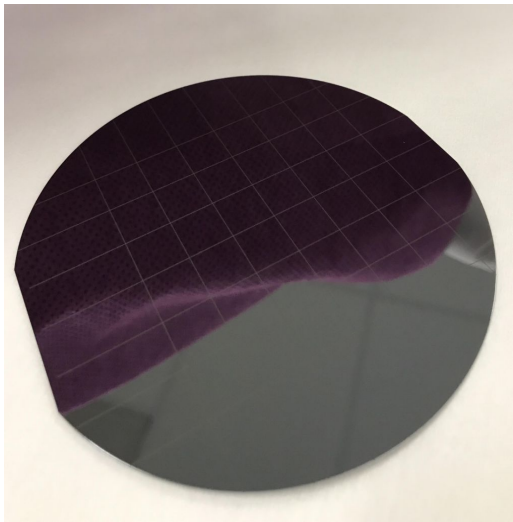


8) Secure with Rectangle Stopper

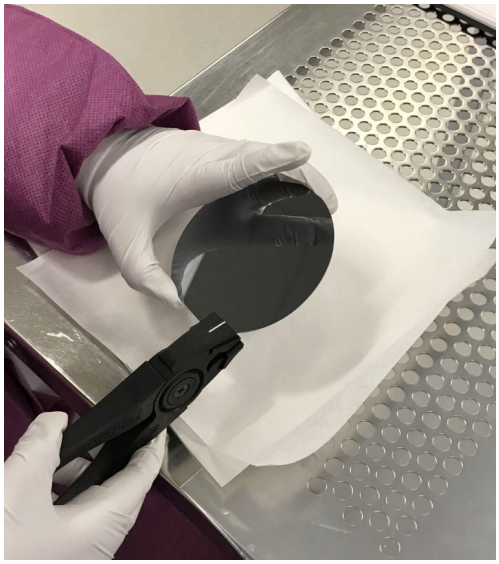


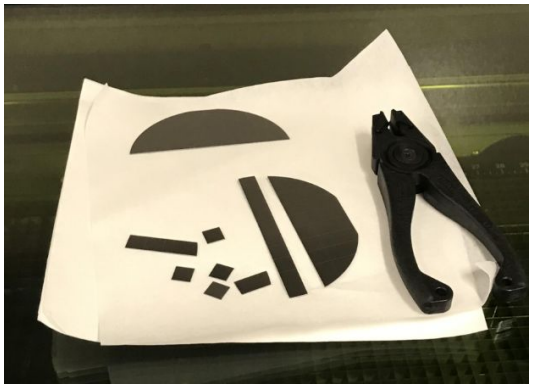
9) Dice with Laser Cutter

a) Fiber Laser: Vector Only: Vector Parameters: 10% Speed, 50% Frequency, 50% Power



10) Use an all plastic wafer cleave plier to dice





11) Transport back to SNF main cleanroom in clean container

12) DI water clean @ wetbench

13) Load die on carrier wafer (made in step 2) with drop of vacuum oil

14) Use dry etcher

a) Lampoly

i) Run standard clean recipe

ii) Si3N4 etch recipe

iii) Run standard clean recipe

b) Drytek2

c) P5000

i) Pre-Clean: Chamber Clean to be run before the Trench Etch Process

ii) P5000 Chamber B: Si3N4 Etch Recipe

iii) Post-Clean Chamber Clean to be run before the Trench Etch Process

iv) Post-Clean2: Clean to Minimize Grass

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

N/A

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

N/A

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

N/A

8.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: